Cryptocurrency Portfolio Analysis

Bitcoin (BTC), Ethereum (ETH), and Solana (SOL)

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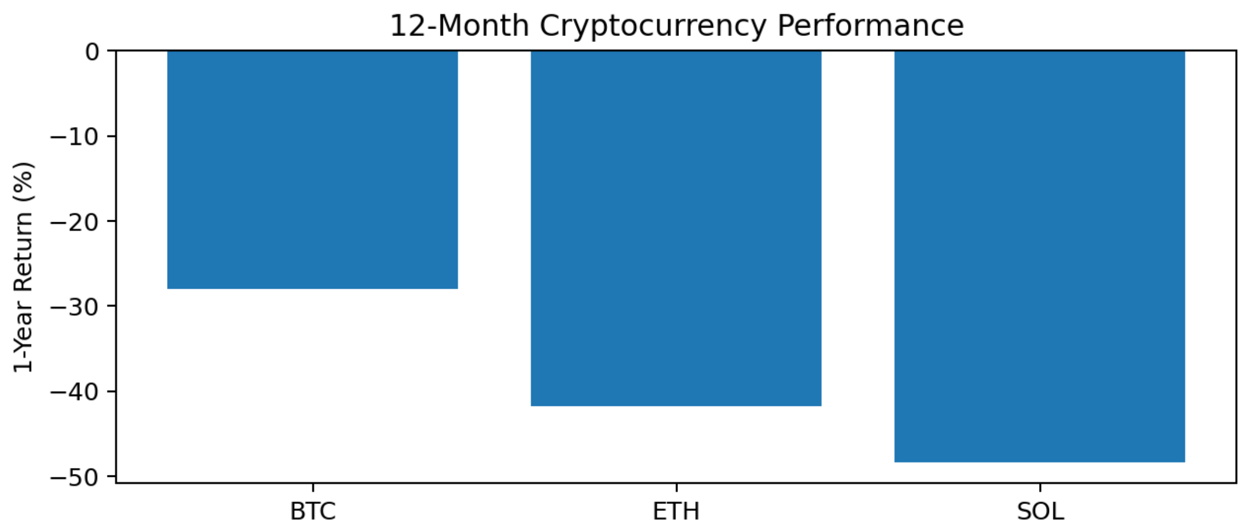
1. Objective and Methodology

This report compares BTC, ETH, and SOL over approximately the latest 365-day period. The analysis focuses on return, volatility, maximum drawdown, Sharpe ratio, Value at Risk (VaR), Expected Shortfall, and diversification characteristics. Risk statistics are based on daily crypto returns and 365-day annualization. The figures are educational and are not personal investment advice.

2. Performance Overview

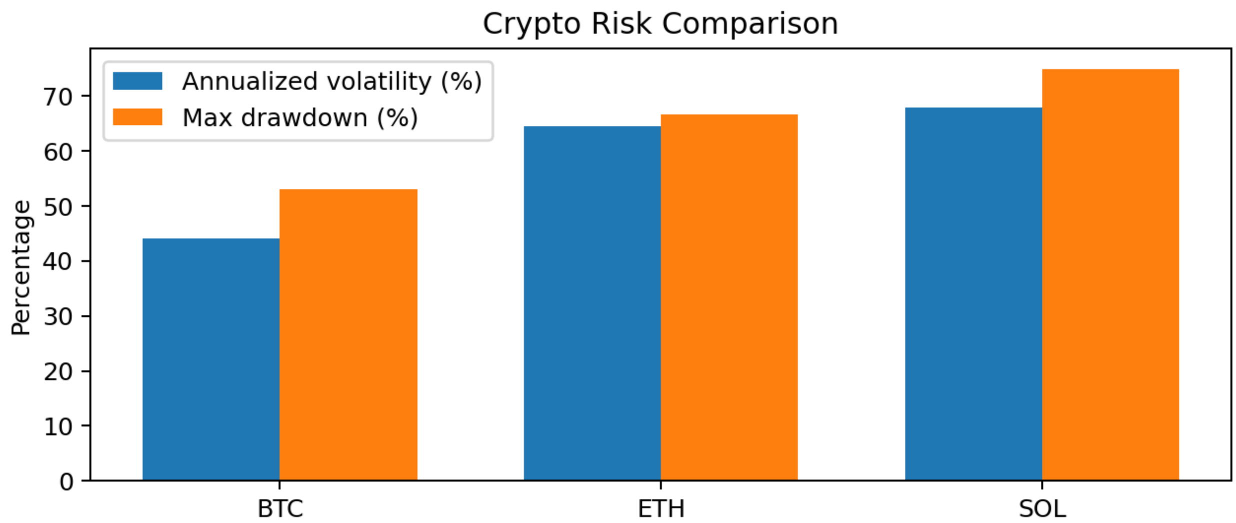
All three assets produced negative one-year returns. Bitcoin was the strongest performer at -28.0%, followed by Ethereum at -41.8% and Solana at -48.4%. This indicates that the period was challenging for the selected crypto assets despite substantial rallies during shorter intervals.



Metric	Bitcoin (BTC)	Ethereum (ETH)	Solana (SOL)
Current price	$79,415.18	$2,489.45	$106.49
Market cap	$1.595T	$303.75B	$62.34B
1-year return	-28.0%	-41.8%	-48.4%
Annualized volatility	44.2%	64.5%	67.9%
Maximum drawdown	-53.0%	-66.7%	-74.9%
Sharpe ratio	-0.61	-0.58	-0.69

3. Risk Analysis

Bitcoin had the lowest annualized volatility (44.2%) and the shallowest maximum drawdown (-53.0%) of the three. Ethereum's volatility was 64.5% with a -66.7% maximum drawdown. Solana had the highest volatility at 67.9% and the deepest drawdown at -74.9%. Thus, SOL had the greatest historical downside and price variability in this one-year comparison.



At the 5% daily-log-return level, BTC's VaR was -3.67% and Expected Shortfall was -5.32%; ETH's were -5.19% and -7.87%; SOL's were -5.81% and -8.41%. Expected Shortfall estimates the average loss on observations beyond the 5% VaR cutoff. These statistics emphasize the heavier downside risk in ETH and especially SOL.

4. Diversification and Correlation

The assets are strongly correlated. BTC and ETH had an average one-year correlation of about 0.89, while BTC and SOL were about 0.85 and ETH and SOL about 0.87. High positive correlation means that holding all three does not eliminate crypto-market risk; they have historically tended to decline together during major market stress.

5. Individual Assessment

Bitcoin: The comparatively lower volatility and smaller drawdown make BTC the least risky of the three in this analysis. Its -28.0% return was also the strongest. Its principal risk remains broad crypto-market and macroeconomic exposure.

Ethereum: ETH delivered a weaker return than BTC and had materially higher volatility. However, it outperformed SOL and had a less negative Sharpe ratio than both BTC and SOL in the selected window. Its risk remains elevated because of large price swings and competition among smart-contract platforms.

Solana: SOL produced the weakest return and highest volatility/drawdown of the three. Its ecosystem and application growth can provide upside potential, but the historical risk profile suggests that position sizing should be more conservative than BTC.

6. Conclusion and Recommendation for a Hypothetical Portfolio

For a hypothetical diversified crypto portfolio, BTC would be the core holding because it had the strongest one-year return and lowest measured volatility/drawdown among the three. ETH could be used as a secondary allocation, while SOL would be treated as the higher-risk satellite allocation. Because correlations are high, diversification among BTC, ETH, and SOL should not be interpreted as protection against a broad crypto-market decline. A disciplined allocation, periodic rebalancing, and a predefined risk limit are therefore important.

References

• Gale Finance, Bitcoin vs Ethereum (1-year analysis, updated 7 Sep 2026).

• Gale Finance, Bitcoin vs Solana (1-year analysis, updated 6 Sep 2026).

• Gale Finance, Ethereum vs Solana (1-year analysis, updated 4 Sep 2026).

- CoinGecko, cryptocurrency market and historical price data, accessed 7 Sep 2026.